

Marine Food Pollutants as a Risk Factor for Hypoinsulinemia and Type 2 Diabetes

Background Some persistent environmental chemicals are suspected of causing an increased risk of type 2 diabetes mellitus, a disease particularly common after age 70. This concern was examined in a cross-sectional study of elderly subjects in a population with elevated contaminant exposures from seafood species high in the food chain. **Methods** Clinical examinations of 713 Faroese residents aged 70-74 years (64% of eligible population) included fasting plasma concentrations of glucose and insulin, and glycosylated hemoglobin. Lifetime exposure to persistent environmental chemicals from pilot whale and other traditional food was estimated from a dietary questionnaire and by analysis of blood samples for polychlorinated biphenyls (PCBs) and related food contaminants. **Results** Septuagenarians with type 2 diabetes or impaired fasting glycemia tended to have higher PCB concentrations and higher past intake of traditional foods, especially during childhood and adolescence. In non-diabetic subjects, the fasting insulin concentration decreased by 7% (95% CI= -12% to -2%) for each doubling of the PCB concentration after adjustment for sex and body mass index at age 20. Conversely, the fasting glucose concentration increased by 6% (-1% to 13%) for each doubling in PCB. Similar associations were seen in subjects without impaired fasting glycemia, while further adjustment for current body mass index and lipid metabolism parameters attenuated some of the associations. **Conclusions** Impaired insulin secretion appears to constitute an important part of the type 2 diabetes pathogenesis associated with exposure to persistent lipophilic food contaminants.